The Role of the IPCC in Climate Science FREE

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Summary

The Intergovernmental Panel on Climate Change (IPCC) consists of about 190 governments that commission assessments performed by the international climate science community to *determine the current state of human knowledge of climate and climate change*. As such, the IPCC does not perform scientific research, but, rather, assesses research in the form of published papers addressing topics in climate science related to climate variability and change. However, as the IPCC assessments have evolved (from the first in 1990 to the sixth in 2021, so far), the IPCC has formed a symbiotic relationship with climate science. Even though the goal of the IPCC is to assess the scientific research that is taking place, its high profile, prestige, and interest from governments that fund climate science research has stimulated and arguably accelerated climate science research. This is particularly relevant for Earth system modeling (including the physical climate system plus the biogeochemical components) that will be addressed here to illustrate the influence of IPCC on climate science. One outcome is that enhanced observations of the Earth system from a number of field campaigns have been funded by countries to gather targeted observations to improve the understanding of crucial processes that need to be represented with fidelity in Earth system models. Governments that fund Earth system modeling research want to have results from their model appear prominently in the IPCC assessments to partially justify the funds being spent on developing, running, and analyzing these models. And just as important as getting a model into the IPCC assessment process are the analyses of the model outputs done by the scientists in the modeling groups and other scientists around the world. The products of this process are the papers describing cutting-edge results that use the models to advance knowledge of climate variability and change. Therefore, model developers are competing with other modeling groups around the world to have the best possible models producing climate simulations that are analyzed to produce papers of the highest quality that are assessed in the IPCC reports. An important part of this process is the international scientific coordination provided by the World Climate Research Programme's Coupled Model Intercomparison Project (CMIP). CMIP involves climate scientists from around the world who volunteer their time to organize CMIP while also developing climate models in their respective countries. CMIP started in the mid-1990s for modeling groups to run exactly the same experiments so the response across the models could be directly compared to quantify uncertainty in their simulations of historical and future climate. Because these climate experiments are, by construction, the current state-of-the-art in climate modeling with the best representation of human understanding of the workings of the climate system, the papers that are written based on those model integrations are of primary interest for the IPCC assessments. CMIP has since evolved to include numerous climate science communities that interface with the modeling groups to perform model intercomparison projects to address various compelling climate science problems. Thus, there is a symbiosis between climate science/modeling, the scientific framework provided by CMIP for coordinated climate change experiments, and the IPCC process that assesses papers that emerge from the scientific research done by scientists who desire their work to be featured in those prestigious IPCC assessments.

Keywords: IPCC, climate science, climate modeling, Earth system modeling, CMIP, WCRP

Subjects: Policy, Politics, and Governance

The Intergovernmental Panel on Climate Change

The Intergovernmental Panel on Climate Change (IPCC) consists of about 190 governments that commission assessments performed by the international climate science community to *determine the current state of human knowledge of climate and climate change*. The scientists are divided into three working groups: Working Group 1 (WG1) deals with the physical basis of climate change; Working Group 2 addresses climate impacts, adaptation, and vulnerability; and Working Group 3 assesses mitigation of climate change. The IPCC was established by the United Nations Environment *<http://www.ipcc.ch/docs/UNEP_GC-14_decision_IPCC_1987.pdf>* Programme (UNEP) *<http://www.ipcc.ch/docs/UNEP_GC-14_decision_IPCC_1987.pdf>* and the World Meteorological Organization (WMO) *<http://www.ipcc.ch/docs/WMO_resolution4_on_IPCC_1988.pdf>* in 1988 to provide the world with a clear scientific view on the current state of knowledge of climate change and its potential environmental and socioeconomic impacts (Bolin, 2007). In the same year, the UN General Assembly endorsed the action by UNEP and WMO in jointly establishing the IPCC *<http://www.ipcc.ch/docs/UNGA43-53.pdf>* (Bolin, 2007).

There are a number of unique features of IPCC assessments. First, they provide an assessment, not a review. This is a key distinction because a review provides a survey of relevant literature, while an assessment does that and then goes on to form conclusions regarding the current state of the science. Second, they are policy relevant, but not policy prescriptive. This is also important because the scientists must focus on assessing the science that may be brought to bear on policy decisions and be very careful not to make policy recommendations. Third, the assessment process is transparent in that each assessment involves two stages of open and extensive international review. Reviewers (ranging from scientists to some qualified members of the public) volunteer to provide comments. Each comment is catalogued, documented, and responded to by the lead authors. For each round of review, a single chapter may receive ~1,500 or more comments. All comments and responses can be traced and are openly available at the end of the process. In this way, each reviewer can track the lead authors' responses to their comments. Fourth, to quantify the certainty of the assessment conclusions, calibrated uncertainty language is applied as a standard part of the WG1 assessments (the other working groups have "confidence" language and other forms of calibrated language).

This calibrated uncertainty language is something unique to the IPCC assessments. The scientists involved with the assessment process have to not only learn what these terms and likelihood of probability ranges mean, but also become comfortable with this language. This is because the calibrated uncertainty language represents a quasi-quantification of how confident the authors are with particular conclusions in their assessment of the science. The language evolved as the IPCC assessments evolved but was established fairly early on with words that represented probabilities that a given assessment statement is well established. These calibrations of assessment certainty range from "virtually certain" (representing a 99% or greater probability of the likelihood of the outcome), to "extremely likely" (95% or greater probability), to "very likely" (greater than 90% probability), to "likely" (more than 66% probability), to "about as likely as not" (ranging from 33% to 66% probability), to "unlikely" (less than 33% probability), to "very unlikely" (less than 10% probability), and "exceptionally unlikely" (less than 1% probability). The shades of certainty represented by this calibrated language are vigorously debated at nearly all levels of the assessment process, starting first among the scientists who are performing the assessment and including the reviewers who provide critical reviews of the chapters and the government delegations who approve the report at the end of the process.

This article will focus mainly on IPCC WG1 climate science, and Earth system modeling in particular, to illustrate the influence of IPCC on climate science.

The Emergence of IPCC Influence on Climate Science and the First Assessment Report

When the first Intergovernmental Panel on Climate Change (IPCC) assessment report was being formulated in the late 1980s, the climate science community had just gone through a mostly U.S.–based assessment as part of the four Department of Energy State-of-the-Art Reports on climate change (e.g., MacCracken & Luther, 1985a, 1985b). Because it had just been formed, the IPCC was not yet known in the climate science community, and it was viewed as a more international version of what the mostly U.S. scientists had just done for the U.S. Department of Energy. An organizational workshop was held at the Geophysical Fluid Dynamics Laboratory (GFDL)/National Oceanic and Atmospheric Administration in Princeton, New Jersey, in 1988 with members of the then relatively small climate change scientific community. A larger workshop involving more climate scientists was convened in Amherst at the University of Massachusetts later in 1988 to come up with a chapter outline and form lead author teams. Leadership was provided by climate scientists from the United Kingdom, as Sir John Houghton, former head of the British Meteorological Office, had taken on the role as first chair of the IPCC Working Group 1 (WG1).

A main focus was on possible future climate change being simulated by the first generation of climate models just starting to be run with realistic geography for multi-decadal time periods. A first lead author meeting was held in December 1989, in Brisbane, Australia, as the draft assessment was being put together. Papers describing trends in observations and other observational evidence for climate change were assessed. Most of the model simulations of the future projections of possible future climate change due to increasing greenhouse gases (most notably CO_2) were limited to a few atmospheric models coupled to non-dynamic slab oceans. These ocean formulations were usually a 50 m deep slab of inert water with no ocean currents but with the capability to absorb and release heat from that 50 m deep layer. In these experiments, CO_2 was instantaneously doubled, and the climate changes were assessed when the model came to equilibrium with the new forcing. These so-called equilibrium climate change experiments usually required about 30 years of model simulation. They provided the first measures of possible future climate change due to the burning of fossil fuels. This measure of climate change was called "equilibrium climate sensitivity" (ECS) and was defined as the global surface temperature warming at equilibrium after CO_2 was doubled. The history and evolution of values of ECS could form an entire article of their own and can be traced back to the seminal work on human-caused climate change by Klaus Hasselmann and Syukuro Manabe that won them the 2021 Nobel Prize in Physics. The possible range of ECS was first estimated to be 1.5°C–4.5°C in the so-called Charney Report in 1979 (National Research Council, 1979). This ECS range was also assessed to be 1.5°C–4.5°C in the First IPCC assessment and has not changed substantively in subsequent assessments.

Meanwhile, the first generation fully coupled atmosphere-ocean general circulation models (AOGCMs) were just beginning to be run. The model experiments run with these models, the first of their kind, included CO_2 increasing at 1% per year, a so-called "transient" climate change experiment. This was designed to approximate the time evolution of what could happen to the climate system over the next 70 years until CO_2 doubled in that idealized experiment.

Results from two of these new models had been published in 1989 and were assessed in the draft that went through international review. One had been developed at the National Center for Atmospheric Research (NCAR) by a group led by Warren Washington and was published in June 1989. Due to limitations of computer time, this model was run for only the first 30 years of a transient CO_2 increase experiment with CO_2 increasing 1% per year linearly, or only about halfway to a doubling of CO_2 (Washington & Meehl, 1989). The second was developed by a modeling group at the GFDL in Princeton led by Syukuro "Suki" Manabe and was published in December 1989 (Stouffer et al., 1989), making it just under the wire to be included in the final draft of the assessment that went out for international review. In this experiment, the model was also run with CO_2 increasing at 1% per year (compound) but the authors had enough computer time to enable them to run the model out to reach the time of CO_2 doubling at about year 70. Both models had grid resolutions that were quite coarse, with model grid points roughly every 500 km or so in atmosphere and ocean, with very simple land surface formulations that allowed a crude calculation of soil moisture and runoff. But even with these early coarse-grid coupled AOGCMs, they showed characteristics of climate change that are still in evidence in current Earth system models (ESMs), namely, land warms faster than ocean and there is greater warming over the Arctic than anywhere else, with less warming in the North Atlantic and Southern Ocean.

The draft assessment report went through international review, and at a final lead author meeting in the United Kingdom in the spring of 1990, future Nobel Prize winner Suki Manabe attended and showed new results that elaborated on his group's 1989 paper and illustrated what he considered an important and vital point. He noted that the global mean surface warming at the time of CO_2 doubling in his group's fully coupled AOGCM was only about 60% of that in an equilibrium CO_2 doubling simulation with an atmosphere coupled to a non-dynamic slab ocean. This lag of warming was due to processes in the dynamical ocean model that absorbed some of the heat from increasing CO_2 in the atmospheric model. Manabe felt this result was very policy relevant given that any policy actions taken to mitigate CO_2-induced climate change at any point in time would be seeing only a fraction of the eventual warming that would occur.

Manabe was keen to have this result included in what he accurately anticipated would be a notable scientific report with a high profile that governments would pay attention to. However, he had just a draft manuscript, and the IPCC rules stated that only papers published in a peer-reviewed journal before the international review could be assessed. Therefore, according to this rule, his new results could not be included. But IPCC WG1 chair Sir John Houghton knew this was an important scientific result with compelling policy implications, and he argued that Manabe's preliminary result should be included in the IPCC First Assessment Report.

After a lot of argument and one lead author resigning in protest, Manabe's result from an unpublished manuscript was included, even though the international review had been completed. The results from that paper were cited in Chapter 6 of the First Assessment Report as "From Manabe (1990) personal communication," and the citation in the reference list at the end of that chapter read, "Manabe, S., 1990: Private communication." This would never have been allowed in subsequent incarnations of the IPCC assessments, but this was the first, and rules were still fluid at that stage. Inclusion of this climate model result from one of the notable pioneers in the field sent a message to the rest of the climate science community. Manabe was taking the IPCC seriously, and he realized that there was the potential to reach governments with the message of human-caused climate change being a real issue for the future.

The First Assessment Report was published in 1990 (IPCC, 1990). Because at that time there was no indication there would be a second assessment report (SAR), the first did not have a number but was titled just "The IPCC Scientific Assessment." It was only after the SAR was published in 1995 that the first was called the First Assessment Report. In any case, shortly after the First Assessment Report appeared, there was a big ramp-up to the Earth Summit in Rio de Janeiro to be held in 1992. This would be the first gathering of government delegations to address the human-caused climate change problem. Subsequent to the publication of the First Assessment Report, the NCAR group had run its 1% CO_2 increase experiment out to the time of CO_2 doubling at about year 70 as GFDL had, and two other modeling groups (the British Met Office in the United Kingdom, and the Max Planck Institute in Germany) had completed similar 1% per year transient CO_2 increase experiments to the time of CO_2 doubling. Prompted in part by this new suite of global coupled models running the same transient climate change experiment, it was decided to formulate an "update" to the WG1 1990 assessment (IPCC, 1992). In addition to these new modeling results that complemented and confirmed the NCAR and GFDL results in the First Assessment Report, there were other new scientific results analyzing observations and other aspects of climate change. These were assessed in a Supplementary Report to the First IPCC assessment (IPCC, 1992). The governments that met in Rio used the First Assessment Report and the Supplement to inform their deliberations.

The geographical pattern of climate change at the time of CO_2 doubling that emerged in all four of the AOGCM experiments with first-generation coupled climate models showed results that subsequent generations of ESMs have simulated for future climate change. As noted earlier, these included land warming faster than ocean, the Arctic warming faster than anywhere else, and the North Atlantic and Southern Ocean warming slower (Figure 1).

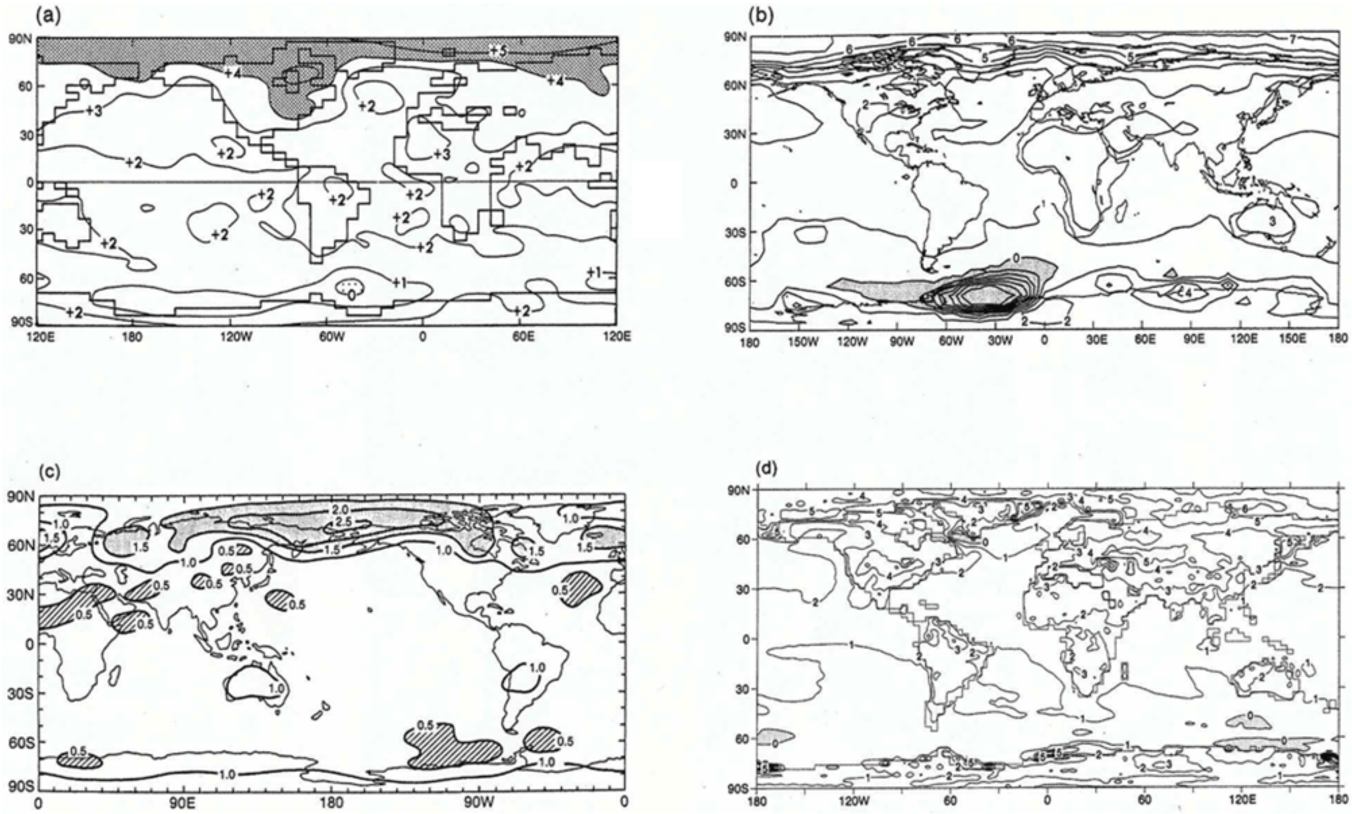

Figure 1. Changes in surface temperature around the time of CO_2 doubling near year 70 in a "transient" experiment with CO_2 increasing at about 1% per year from the first four global coupled climate models in 1992 update to the Intergovernmental Panel on Climate Change First Assessment Report, (a) Geophysical Fluid Dynamics Laboratory (GFDL), (b) Max Planck Institute (MPI), (c) National Center for Atmospheric Research (NCAR), and (d) United Kingdom Meteorological Office (UKMO) (afterIPCC, 1992).

The fact that this basic pattern of surface temperature response to an increase of CO_2 has not changed in subsequent generations of more sophisticated and complex ESMs is reassuring. This indicates that very fundamental processes in the climate system are affected by increasing CO_2 and could be captured in even the first generation of relatively crude climate models.

As all this was going on in the IPCC realm, on the science side the World Climate Research Programme (WCRP), under the auspices of the United Nations, could see that the emergence of multiple global coupled climate models was creating a scientific need for coordination among the groups to run similar experiments so the model results could be more readily analyzed and compared. For example, some modeling centers were running 1% per year CO_2 increasing linearly, and some with a compound increase. Thus, direct comparison was complicated by groups not running exactly the same experiment. Therefore, in 1990, the WCRP formed the Steering Group on Global Coupled Models chaired by Larry Gates. This committee had representatives from the modeling groups who were performing climate change experiments with global coupled climate models that were the most credible tools to simulate the time-evolving response of the climate system to increasing CO_2.

As the climate science community organized to meet that scientific challenge, the governments of the IPCC were pleased with the 1990 assessment and the 1992 update that informed the first formal international recognition of the human-caused climate change problem by the governments of the world at Rio. This appeared to open the door to negotiations to take action to address the problem. The IPCC decided that to inform these ongoing negotiations, periodic scientific assessments should be performed by the international climate science community. Therefore, a SAR to be published in 1996 (IPCC, 1996) was put in motion. As this was going on, the climate science community represented by WCRP decided that as more modeling groups were entering the high stakes activity of attempting to estimate the time evolution of future climate with global coupled climate models, a more formal structure for experimentation and coordination should be devised. An organizational workshop at Scripps Institution of Oceanography in 1994 introduced the possibility of a formal model intercomparison activity whereby modeling groups would run the same experiments (initially a control run and a 1% per year CO_2 compound increase to time of CO_2 doubling at about year 70 in such an experiment). What grew out of that workshop was the Coupled Model Intercomparison Project (CMIP). This was modeled after a previous atmospheric model intercomparison activity called the Atmospheric Model Intercomparison Project (AMIP) that was also organized by Larry Gates (e.g., Gates et al., 1999). In the AMIP experiment design, atmospheric models were run with time-evolving observed sea surface temperatures and sea ice from 1979 to near present. The resulting simulations, performed in the same way by multiple modeling groups, could be directly compared to identify systematic errors and to document model fidelity in simulating observed time-evolving climate. The original idea for CMIP was to expand the AMIP concept to include fully coupled ocean-atmosphere climate models performing simulations of present-day (e.g. simulations of 20th century climate with observed increases of greenhouse gases and other forcings) and future climate. The formation of CMIP had the 1996 IPCC assessment in mind but was motivated by the scientific question of what future climate would look like in a world with increased CO_2. Thus began the symbiotic relationship between the climate science community and the IPCC assessment process.

The Second IPCC Assessment Report and the Rise of the Climate Change Denier Community

As more modeling groups from more countries began to formulate climate models to study climate change, the Coupled Model Intercomparison Project organized a mechanism for model output to be collected in a central location at the Program for Climate Model Diagnosis and Intercomparison (PCMDI) that had formed under the leadership of Larry Gates (Potter , 2011). Scientific transparency dictated that climate change output from these cutting-edge models should be widely available for analysis by the climate science community. PCMDI offered that mechanism whereby climate scientists could request model data in the same format that PCMDI had collected from the modeling groups for ready analysis. This facilitated an acceleration of scientific papers with climate change results from analyses of these models.

As the research infrastructure was being put together on the science side, a request to the climate science community arrived from the Intergovernmental Panel on Climate Change (IPCC) to perform a second assessment report (SAR). As the assessment process got underway and scientific papers published since the first report were being assessed by lead author teams formed from the international climate science community, the Steering Group on Global Coupled Models collected output from 1% per year CO_2 increase experiments from 10 modeling groups in four countries, and these results were published in the SAR (Figure 2).

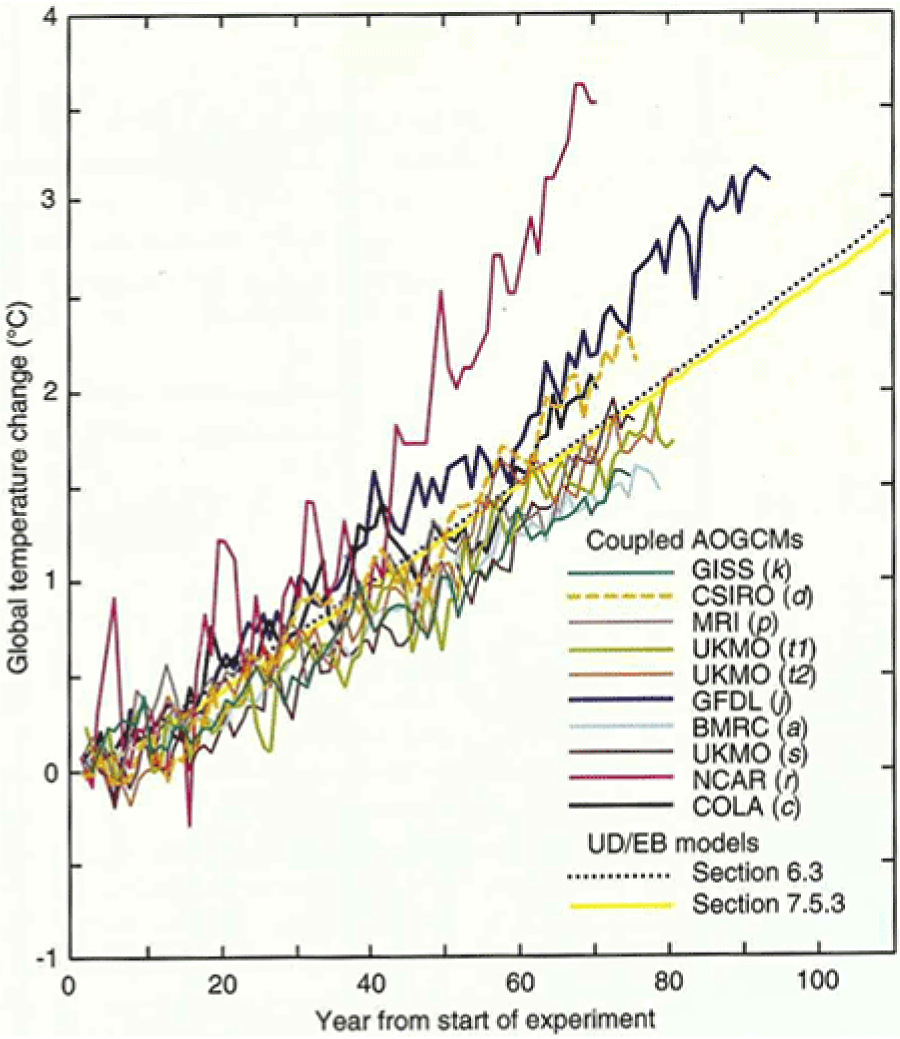

Figure 2. Time evolution of global mean surface temperature in atmosphere-ocean general circulation model simulations from 1% CO_2 increase experiments performed by modeling groups listed in the legend, collected by Steering Group on Global Coupled Model and assessed in the second assessment report (Fig. 6.4 in IPCC, 1996).

The results from this larger group of first-generation climate models were in rough agreement with the four models from the first assessment report, though the spread of the models was notable and highlighted the uncertainty in the magnitude of possible future climate change, an issue that is still prevalent in the early 21st century.

Another development on the climate science side that had profound ramifications for the IPCC assessment process was the emergence of a new area of climate change research called "climate change detection and attribution." This area of research first asked whether a certain climate change signal could be detected above the background weather noise of the climate system. If such a climate change was detected in that way using various statistical methods employing observations and climate model data, then the question was asked whether such a climate change could be attributed to a cause, whether human or natural. Thus, climate change detection/attribution brings together observations and models in a meaningful way such that detection involves observations and attribution relies on climate models. For the latter, the so-called anthropogenic fingerprint of CO_2-induced climate change came from the atmosphere-ocean general circulation model data being produced by the CO_2 increase experiments being run by the international modeling groups and collected at PCMDI. This led to papers being written on this topic and assessed in the SAR. The lead authors' assessment of those papers led to the first "smoking gun" statement made by the climate science community that tangibly linked human activity to climate change. This statement has become one of the scientific icons in climate science history whose implications still reverberate: "The balance of evidence suggests a discernible human influence on global climate."

No one involved with the SAR had any idea the reaction that single sentence would trigger. The message received by the energy industry and in some political circles was that climate science now presented a perceived tangible threat to a business model that depended on unimpeded burning of fossil fuels, and thus energized the nascent climate change denier movement. The weight of credibility associated with this IPCC scientific assessment, which was commissioned by the governments of the world, made this threat immediate and real and could threaten businesses if governments decided to reduce emissions from fossil fuel burning.

The reaction was prompt and dramatic. Since the fossil fuel industry and their allies knew they could not challenge the science, they tried to discredit the report by finding administrative flaws in the IPCC assessment process itself. In particular, the final IPCC plenary, where government delegations are present to go over the Summary for Policymakers word for word, produces wording clarifications during the course of the plenary that are discussed by the delegations and scientists who wrote the report. The goal is not to change the message, but to clarify meaning. These changes agreed to in the plenary by the governments and the scientists are then incorporated in a final version of the Summary for Policymakers after the plenary. The climate change deniers seized on the changes that emerged in the final report after the plenary as "unauthorized." This, of course, was wrong but didn't matter because what was presented was a loophole through which they could attack the credibility of the report. The focus was turned on the Coordinating Lead Author of the chapter where that smoking-gun sentence originated, Ben Santer. They then applied the political technique of character assassination to destroy his credibility by questioning his ethics in making such "unauthorized changes." Santer was pilloried in the *Wall Street Journal* <https://www.wsj.com/articles/SB837026462152576000> and in many other media outlets. Because this was the first time Santer and the rest of the climate scientists who had put together the SAR had ever encountered this tactic, they naively assumed that it was all a misunderstanding, that if the climate science behind the smoking-gun statement was carefully explained, all would be well. What the scientists did not understand was that the deniers knew the science very well, but they were employing political tactics to discredit Santer, and thus the rest of the report by association *<https://www.wsj.com/articles/SB834512411338954000>*. Their goal was for governments not to act on that information and not to form policies to limit the burning of fossil fuels.

The IPCC subsequently instituted review editors, at least two per chapter, that could function in a similar way as do editors of scientific journals. They would stand between critics and the lead authors to address procedural or other questions related to the IPCC assessment process and supervise the international review mechanisms. However, the furor that sprung from the SAR was a wake-up call to the climate science community, who now realized the high stakes that were involved in the study of the science of climate change. There was a realization that climate change research must accelerate to produce very high-quality science to address the human-caused climate change problem. The rise of the denier movement meant that climate science would be highly scrutinized and thus must be of impeccable quality.

New Future Climate Change Scenarios, the Advent of the Coupled Model Intercomparison Project (CMIP), and the IPCC Third Assessment Report (TAR)

At the heart of the climate change issue was how to understand not only the climate change the earth had already experienced, but also what the future may hold. For the former, a combination of analyses of observations and climate model simulations of "historical" climate was required. For the latter, those same climate models, intending to gain credibility by accurately simulating the variability and spatial characteristics of the present-day climate, would be run with possible future increases in greenhouse gases. This was a scientific challenge to the climate modeling community, and, starting in 1995, the response was a set of coordinated experiments that became part of the first two phases of the Coupled Model Intercomparison Project (CMIP). CMIP1 included climate model simulations of present-day climate (21 models from nine countries), and CMIP2 formed a multi-model data set of 1% per year compound CO_2 increase experiments out to year 70 when CO_2 doubled in those experiments (18 models from eight countries; Meehl et al., 1997).

These experiments were begun before the Intergovernmental Panel on Climate Change (IPCC) Third Assessment Report (TAR) was organized. They were motivated by science objectives that included, for CMIP1, a documentation of systematic mean climate simulation errors of global coupled models in atmosphere, ocean, and cryosphere; quantification of the effects of flux adjustment (a technique to correct for systematic errors in the coupled model simulations, e.g., Cubasch, 1991; Manabe & Stouffer, 1988) on the coupled simulations of mean climate and climate variability; and a documentation of the features of simulated climate system variability on a variety of time and space scales. For CMIP2, science goals included documenting the mean response of the dynamically coupled climate system to a transient increase of CO_2 in the models near the time of CO_2 doubling; quantifying the effects of flux adjustment on climate sensitivity in the coupled climate simulations; and documenting the features of the simulated time-evolving climate system response to gradually increasing CO_2. These simulations were run and then subsequently assessed as part of the TAR.

At the same time, observational studies were being performed to quantify climate changes already experienced, and paleoclimate model simulations of past climates with conditions much different from present-day provided insights into processes and mechanisms in the climate system that could function in a variety of temperature regimes (e.g., Kutzbach, 1981). These studies provided credibility for the future climate projections from the models.

Even as the IPCC second assessment report process had gotten underway in the early 1990s, the integrated assessment modeling community in Working Group 3 was coming up with more plausible emission scenarios than the idealized 1% per year increase experiments that had been run and analyzed by the climate modeling groups up to that time. That community requested that the climate modeling community run many of these scenarios they had devised, but computer time limitations meant that only a few modeling groups could run a few of the new scenarios (these included the IS92a and the Special Report on Emission Scenarios (SRES) scenarios;IPCC, 2001). The idea was that producing plausible outcomes for 21st-century climate in terms of emission scenarios that would span a range of what might actually happen would provide greater scientific credibility to the future climate projection simulations being produced by the global coupled climate models.

The CMIP1 and CMIP2 climate model simulations were performed and analyzed using the new emission scenarios that were just becoming available. The resulting scientific publications were assessed in the TAR that was published in 2001 (IPCC, 2001). Figure 3 shows examples of the projected surface air temperature increases from the 1% experiments compared to emission scenario simulations for the end of the 21st century from the IS92a and SRES scenarios.

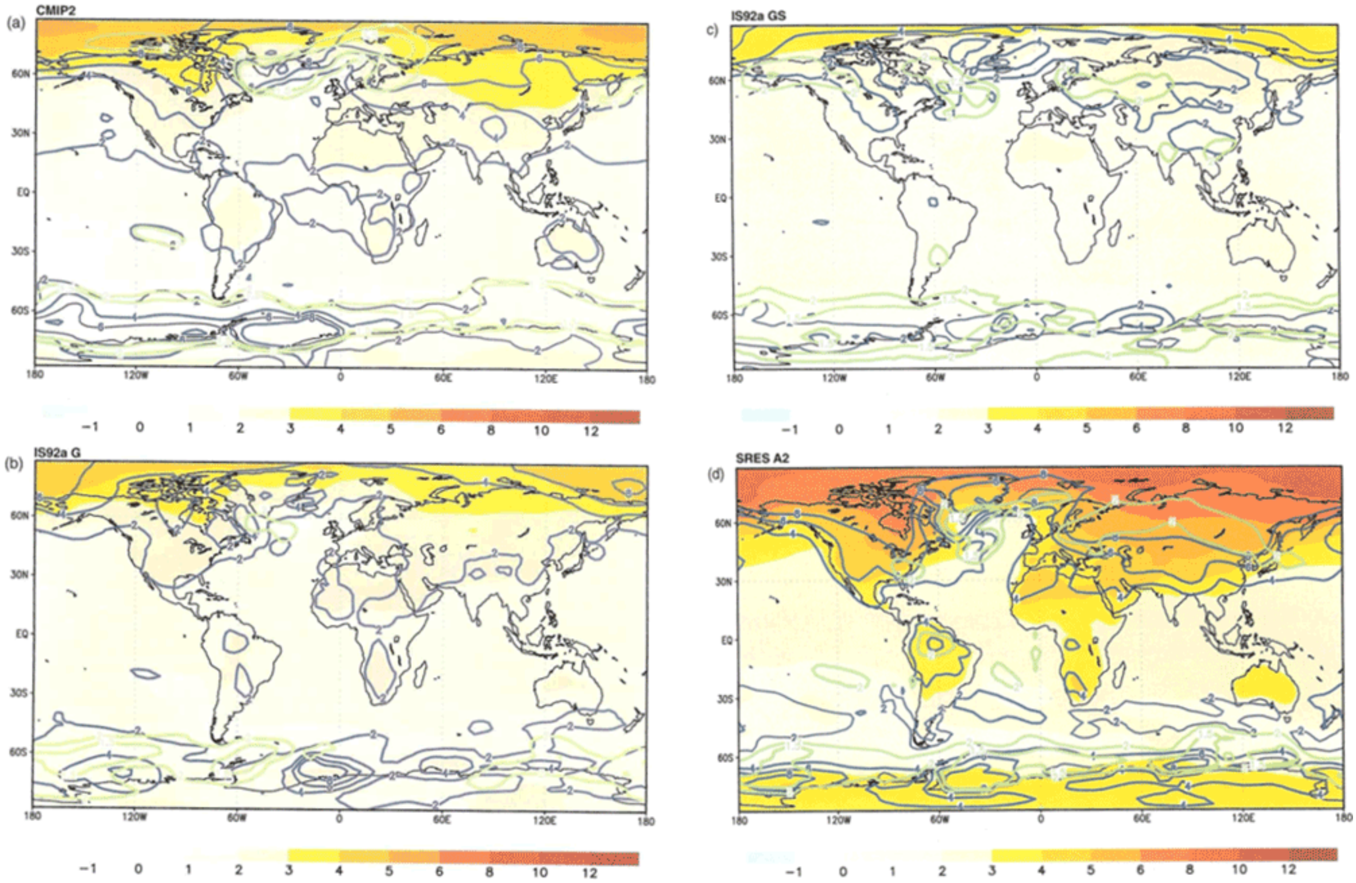

Figure 3. Surface temperature anomalies for increases of CO_2 for (a) Coupled Model Intercomparison Project 2 models with 1% per year CO_2 increase around the time of CO_2 doubling at year 70; (b) end of 21st-century temperature changes from the IS92a G scenario; (c) same as (b) except for IS92a GS scenario; (d) same as (b) except for SRES A2 scenario.

Source: From the Third Assessment Report, IPCC (2001).

The basic patterns of change were similar to the first generation of models depicted in the first assessment report and the 1992 update, but the magnitude of the warming patterns depended on the size of the forcing (i.e., the amount of increasing greenhouse gases in each respective scenario). However, having an actual timeline extending to the end of the 21^{st} century in the IS92a and SRES scenarios provided governments with a more tangible idea of how big the climate changes could be, and when they might happen.

Another science question arose as part of the IPCC process for the TAR related to weather and climate extremes. Scientists in Working Group 2 (WG2) noted that most of the impacts on human society and ecosystems are not related to changes in mean climate but rather to changes in extremes (e.g., heat waves, floods, drought, etc.). The WG2 scientists asked the climate modelers and climate scientists in Working Group 1 (WG1) why there were not more papers written on possible future changes of extremes that they could use in WG2 to inform possible future impacts of those extremes. This became a central issue when extremes were assessed and there were only few studies that existed to enable an assessment to be made (IPCC, 2001).

It turned out that this was a case of WG1 scientists being primarily concerned with the simulation of mean climate and associated changes. But now the climate modelers and scientists in WG1 were motivated to start looking at extremes in the models and observations, and that research has become a central focus of climate science research. A seminal Aspen Global Change Institute (AGCI) session on extremes that arose from the discussions that started in the TAR assessment process outlined a research strategy addressing extremes that set the climate science community in a new direction (Easterling et al., 2000; Meehl et al., 2000).

Another Phase of the Coupled Model Intercomparison Project (CMIP) Contributes to the Fourth IPCC Assessment Report (AR4)

Spurred on by the rapid development of a new generation of atmosphere-ocean general circulation models and stimulated by a number of scientific research questions that arose during the Third Assessment Report (TAR) process, a new phase of the Coupled Model Intercomparison Project (CMIP), CMIP3, was organized in 2003, well before the announcement that there would be a fourth Intergovernmental Panel on Climate Change (IPCC) Assessment Report, the AR4. A particular scientific focus in CMIP3 was the concept of "climate change commitment." This means that even if CO_2 concentrations could be stabilized immediately, there is a certain amount of climate change already "in the pipeline" due to the thermal inertia of the climate system (Meehl et al., 2005). An important aspect of this research was that even if greenhouse gas concentrations could be stabilized, it would take a number of years for globally averaged surface temperatures to level out, but sea level would continue to rise for more than a hundred years. One of the aims of CMIP3 was to better quantify how much committed climate change was in the system. This topic was of great scientific interest because the response of the deep ocean and other slower-acting parts of the climate system needed to be studied. But this topic also had great policy relevance because the idea that the climate benefits of emission reductions would not be immediate was an important concept for governments to grasp.

With the introduction of more realistic future emission scenarios assessed in the TAR, CMIP3 set out to perform scientific experiments in the multi-model context using a new generation of plausible future emission scenarios, the Representative Concentration Pathways (RCPs; IPCC, 2007). The CMIP3 experiments of historical and future climate with the RCP scenarios, as well as other experiments to address the climate change commitment issue, would be performed in the same way by the international modeling groups. These included a 21st-century simulation with SRES A2 to 2100; a 20th-century simulation to year 2000, then fixing all concentrations at year 2000 values and running to 2100 (CO_2 ~ 360 ppm); a 21st-century simulation with SRES A1B to 2100, then fixing all concentrations at year 2100 values to 2200 (CO_2 ~ 720 ppm; running one member to 2300); and a 21st-century simulation with SRES B1 to 2100, then fixing all concentrations at year 2100 values to 2200 (CO_2 ~ 550 ppm; running one member to 2300). The first of these experiments would become a standard future climate change experiment in CMIP3, with variants depending on new scenarios in all subsequent phases of CMIP. As before, the

Program for Climate Model Diagnosis and Intercomparison (PCMDI) agreed to collect, archive, and distribute the multi-model data to what ended up being thousands of climate scientists from around the world.

What transpired was a set of unprecedented, coordinated climate change experiments from 16 groups (11 countries) and 23 models collected at PCMDI (31 terabytes of model data), openly available, accessed by over 1,200 scientists (Figure 4; Meehl et al., 2007).

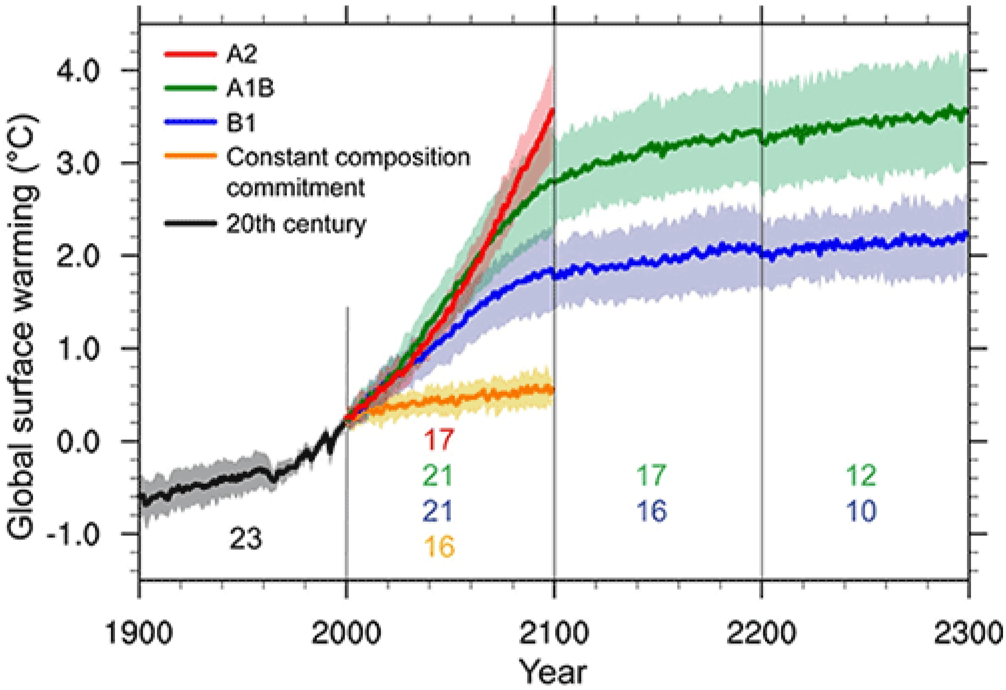

Figure 4. Global mean surface temperature time series for the Coupled Model Intercomparison Project 3 models, historical (black line) mean and range, and for the 21st century from SRES scenarios A2, A1B, and B1; the green and blue lines beyond 2100 indicate further warming if CO_2 concentrations are held fixed at 2100 values in the A1B and B1 scenarios, respectively; the numbers below the time series lines indicate the number of models represented in each time series, and the shading indicates the range across the models (IPCC, 2001).

This was a watershed moment for climate science, as the results from the state-of-the-art climate models from international modeling groups were readily accessible by scientists and students from all over the world for analysis, and hundreds of papers emerged as a consequence. Additionally, climate modeling became transparent because all the model data was openly available for scrutiny, and nothing was hidden from anyone, including the denier community. CMIP3 represented the start of the modern era of open access to multi-model climate data via the Internet and was the foundation for the climate change assessment in the AR4 (IPCC, 2007).

Results from the multi-model data set showed that committed warming averaged 0.1°C per decade for the first 2 decades of the 21st century. Across all scenarios, the average warming was 0.2°C per decade for that time period (compared to a recent observed trend ofnearly the same value).

A contentious scientific point in the AR4 that received much discussion among the IPCC lead authors and elicited scrutiny and criticism from a variety of climate scientists was the assessment of the magnitude of possible future sea level rise. This pointed to a science-IPCC interface issue whereby the lead authors could assess only published scientific papers on estimated future sea level rise. A vast majority of those papers analyzed the CMIP3 multi-model data set and could quantify sea level rise in the models solely due to thermal expansion (as sea water warms, it expands, thus raising global sea levels). None of the models included possible melt from the ice sheets of Greenland and/or Antarctica. There was widespread speculation that those ice sheets could destabilize as warming continued and thus dump huge amounts of meltwater into the global oceans, possibly raising sea level to magnitudes about twice that of thermal expansion. Thus, a thermal expansion of about 0.5 m by 2100 could result in a sea level rise of 1.0 m or more. The ramifications for coastal communities and infrastructure were profound, but the lead authors could assess papers published up to that time documenting only the thermal expansion component. A caveat accompanied the assessment indicating that contributions from melting ice sheets could add to higher sea level rise totals, but there could be no quantitative assessment.

It was clear to the climate science community that interactive ice sheets needed to be included in the climate models, and work along those lines was underway at the time of the AR4, but there were as yet no published results. Prominent mainstream climate scientists criticized the AR4 for not raising a stronger alarm that sea level rise of a meter or more by 2100 could occur. However, the scientific literature published at that time could not support such statements. This was a case of the IPCC assessment being criticized not by the climate change deniers who typically claimed that the assessments were too extreme, but by mainstream climate scientists who thought the report's conclusions were too conservative. Subsequent research that was assessed in the AR6 showed the latter was the case (IPCC, 2021).

In spite of the uncertainty in the possible magnitudes of future sea level rise and other uncertainties inherent in the science, the IPCC AR4 was widely considered to be a definitive statement of the climate change problem at the time it was published in 2007 (IPCC, 2007), and there was a call to action for governments to do something about climate change. In particular, the "smoking-gun" statement linking human activity to climate change that began in the second assessment report, and was subsequently strengthened in the TAR ("There is new and stronger evidence that most of the warming observed over the last 50 years is attributable to human activities"), was even stronger in the AR4: "Most of the observed increase in globally averaged temperatures since the mid-20th century is very likely due to the observed increase in anthropogenic greenhouse gas concentrations." There was widespread media coverage of the release of the AR4, and a growing sense of optimism in the climate science community that the IPCC process could actually produce action at the government level.

However, continuing the campaign to discredit the science that began after the publication of the TAR, the active denier movement focused renewed attacks on the AR4. In 2009, thousands of emails were stolen from a server ("climate-gate"), and several AR4 lead authors' emails were cited out of context to try to ruin those scientists' credibility, with the goal of discrediting the science in the IPCC AR4. Multiple subsequent investigations in the United States and United Kingdom cleared those scientists of any wrongdoing, and the IPCC AR4 science withstood those attacks (Reay, 2010).

The reports from all three working groups were scrutinized by the denier community to look for errors that could discredit the science in the AR4. Two minor errors were found. Both were in the Working Group 2 report, one regarding the magnitude of Himalayan glacier melt (though this information also appeared in the Working Group 1 report and was correct there), and another on the area of the Netherlands below sea level. These errors were corrected, and a better errata procedure was instituted for the AR5.

A Paradigm Shift in Climate Science After the IPCC Fourth Assessment Report (AR4): The Coupled Model Intercomparison Project phase 5 (CMIP5) and the IPCC AR5

Late in the Coupled Model Intercomparison Project phase 3 (CMIP3) during the Intergovernmental Panel on Climate Change (IPCC) AR4 process, it became clear that a profound paradigm shift for climate change science was about to happen. The year of the publication of the AR4 in 2007 was the culmination of coordinated modeling research in CMIP so far and signaled the end of 20 years of relatively coarse grid atmosphere-ocean general circulation models (AOGCMs) run with non-mitigation scenarios. However, the science was evolving, and a new paradigm was emerging. Elements of this new paradigm included a new field in climate science called "decadal prediction," in which climate models were initialized with observations to predict near-term climate variability change on decadal timescales. Additionally, the traditional AOGCMs with components of atmosphere, ocean, land, and sea ice were evolving to first-generation Earth system models (ESMs) with a coupled carbon cycle to study longer term feedbacks past mid-century with new mitigation scenarios, as well as interactive chemistry and aerosols. Because the ESMs were including more components that involved more science disciplines, new tangible linkages throughout the climate science community were required to advance the science.

It was clear to the community that this new paradigm needed a next phase of CMIP to be able to better address these new science problems. To formulate an experiment design to address these new science questions, an Aspen Global Change Institute (AGCI) session in summer of 2006 was convened with about 25 attendees who represented the new set of research communities who were now actively contributing to ESMs to better address the future climate change problem. These participants included climate modelers, chemistry and aerosol modelers, land surface modelers, biogeochemistry modelers, integrated assessment modelers (representing the science assessed by the IPCC Working Group 3 [WG3]), and impacts, adaptation, and vulnerability (IAV) researchers (representing the science assessed by the IPCC Working Group 2 [WG2]). Typically, Working Group 1 (WG1) used the latest state of science models, and WG2 and WG3 used previous model versions. Thus, there was a lag between what version was analyzed in WG2 and WG3 compared to WG1, and this dictated that good communication between the working groups would become essential.

A number of seminal conclusions emerged from that workshop that formed the next phase of CMIP, which would be termed CMIP5. For the first time, the future climate change problem was divided into near-term and long-term timescales, reflecting a shift of the science with the emergence of decadal climate prediction and the needs of the stakeholder community for near-term climate change information. The AGCI session essentially launched the field of decadal climate prediction as a new area of climate science. It also marked the first time ESM experiments were included in a CMIP phase, reflecting the rise of carbon cycle components being included as standard components in AOGCMs. There was a recognized need to connect the Earth System Modeling Community with the integrated assessment modeling (IAM) community in planning a CMIP phase. This was because the IAM community was producing new future emission scenarios, the Representative Concentration Pathway (RCP) scenarios, to be run in the ESMs, and these communities needed to coordinate the science to make that happen effectively. These new RCP scenarios included the concept of how stabilization of climate change could occur (Hibbard et al., 2007). Additionally, coordinated idealized climate model experiments to promote understanding of the climate system were formulated for inclusion in CMIP5. Subsequently, growing directly from the 2006 AGCI session that defined the decadal climate prediction problem, a 2008 AGCI session formulated the first-ever coordinated set of decadal climate prediction experiments that were included in the CMIP5 experimental design (Meehl et al., 2009).

The proposed CMIP5 experimental design was circulated throughout the climate science community for comment and was approved in 2008 by the CMIP Panel, which was part of the World Climate Research Programme (WCRP) Working Group on Coupled Models (WGCM). The timeline for CMIP5 was set before the formulation of the IPCC AR5, and was to run for 5 years, nominally to 2013 (Taylor et al., 2012). As the modeling community set to work with a new generation of ESMs to run the historical and 21st-century RCP scenarios, the IPCC AR5 was announced with a timeline that happened to almost parallel that of CMIP5. Thus, while the science in the CMIP phases, including CMIP5, was always independent of the IPCC assessments, the timelines of the CMIP phases and IPCC assessments roughly mirrored each other. This was because the state-of-the-science results emerging from the model experiments in CMIP, as well as from other areas of climate science, were recognized by the IPCC as being ready to be assessed roughly every 5–6 years.

The IPCC AR5 for WG1 was completed in 2013 (IPCC, 2013). With more models in the CMIP5 database, there was a better quantification of the structural uncertainty for future climate change with the new RCP scenarios that included stabilization scenarios such as RCP2.6 that could be contrasted with higher forcing scenarios with little mitigation such as RCP8.5 (Figure 5a).

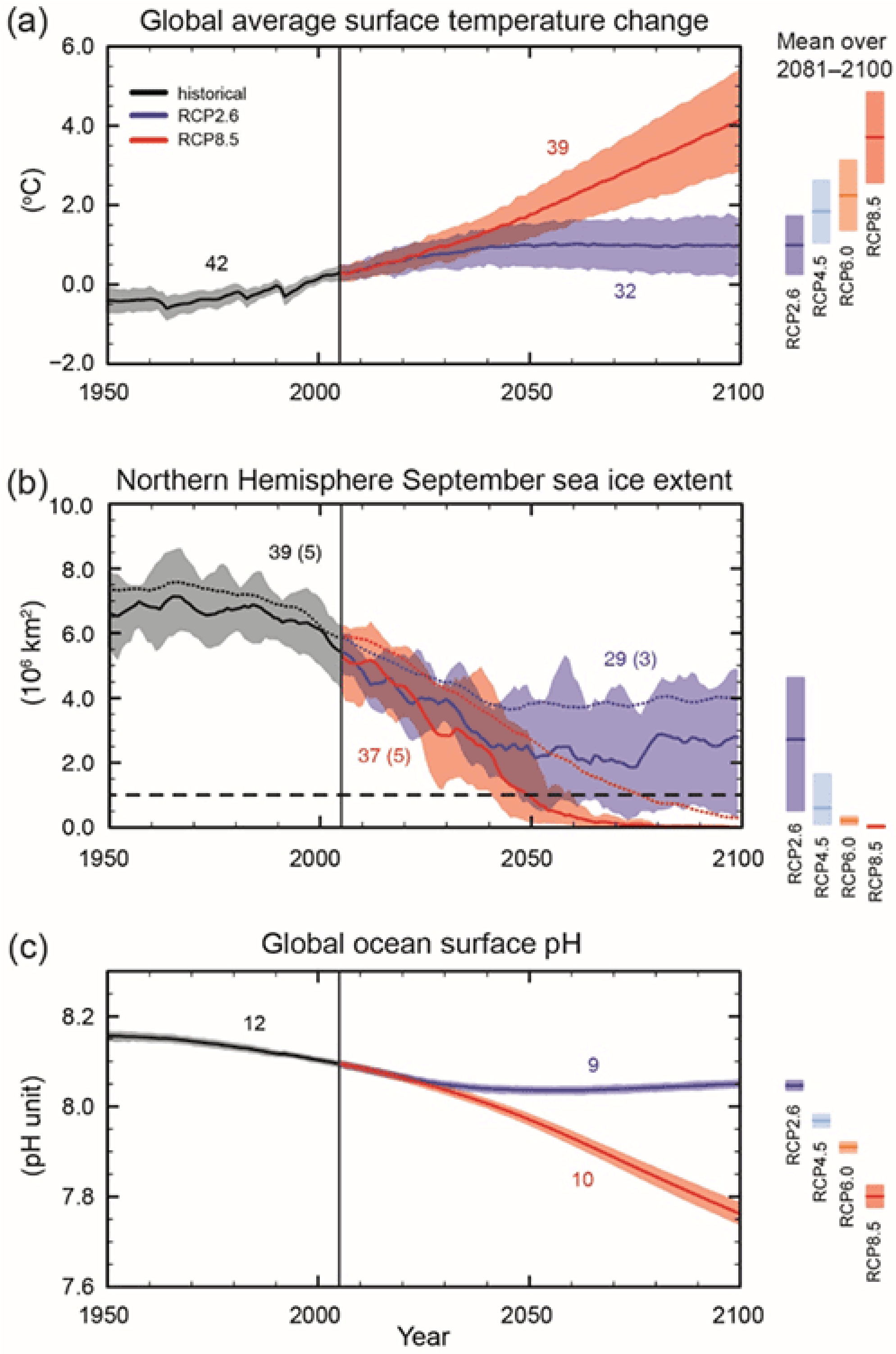

Figure 5. (a) Similar to Figure 4 except for the Coupled Model Intercomparison Project 5 models assessed in the AR5; time series for Representative Concentration Pathway (RCP)2.6 and RCP8.5, the lowest and highest scenarios, are shown with mean and ranges for all scenarios at right; (b) five models that performed best simulating 20th-century Arctic Sea ice extent and thickness estimate an ice-free Arctic in September (defined as extent less than about 1 million square kilometers) occurring roughly in 2050 in the high forcing scenario RCP8.5 (solid red line); (c) global ocean surface pH indicating greater ocean acidification for the high forcing scenario RCP8.5 (red), and stabilized acidification for RCP2.6 (IPCC, 2013).

The idea that some plausible emission scenarios could lead to a stabilized climate in which global surface temperatures were no longer increasing was a key contribution to the policy negotiations that would take place after the AR5 and result in the climate change targets that emerged from the Paris Agreement of 2015 (UN Framework Convention on Climate Change, 2016). In that international agreement, the countries of the world pledged to keep global warming (relative to preindustrial levels) well below 2°C, and ideally less than 1.5°C.

Another new scientific concept that appeared in the AR5 was the idea that a subset of the models that best simulated observations in the 20th century could be used to more credibly estimate possible future climate change. For example, a subset of models that best simulated the extent and thickness of Arctic Sea ice over the latter part of the 20th century were used to estimate when the Arctic could become ice free in September (Figure 5b). Another first for the IPCC AR5 was the standard use of ESMs that included interactive biogeochemistry. This allowed the calculation of possible future pathways of ocean acidification across the future emission scenarios for the first time in an IPCC assessment (Figure 5c).

During the period of the IPCC AR5 in the early 21st century, the rate of warming of global mean surface temperatures slowed down (sometimes referred to as the "hiatus"; Fyfe et al., 2016). Papers addressing this slowdown were assessed, and the conclusion in the AR5 was that much of the slowdown was due to naturally occurring internal variability, associated in particular with the negative phase of the Interdecadal Pacific Oscillation in the tropical Pacific, along with some contributions from a collection of moderate-sized volcanoes in the early 21st century (IPCC, 2013). The slowdown became particularly relevant for assessing possible near-term warming. Climate models applied for the new near-term decadal climate prediction time horizon were initialized for 10-year predictions during the slowdown and consequently showed somewhat less warming in the near term than the traditional uninitialized free-running models (Figure 6).

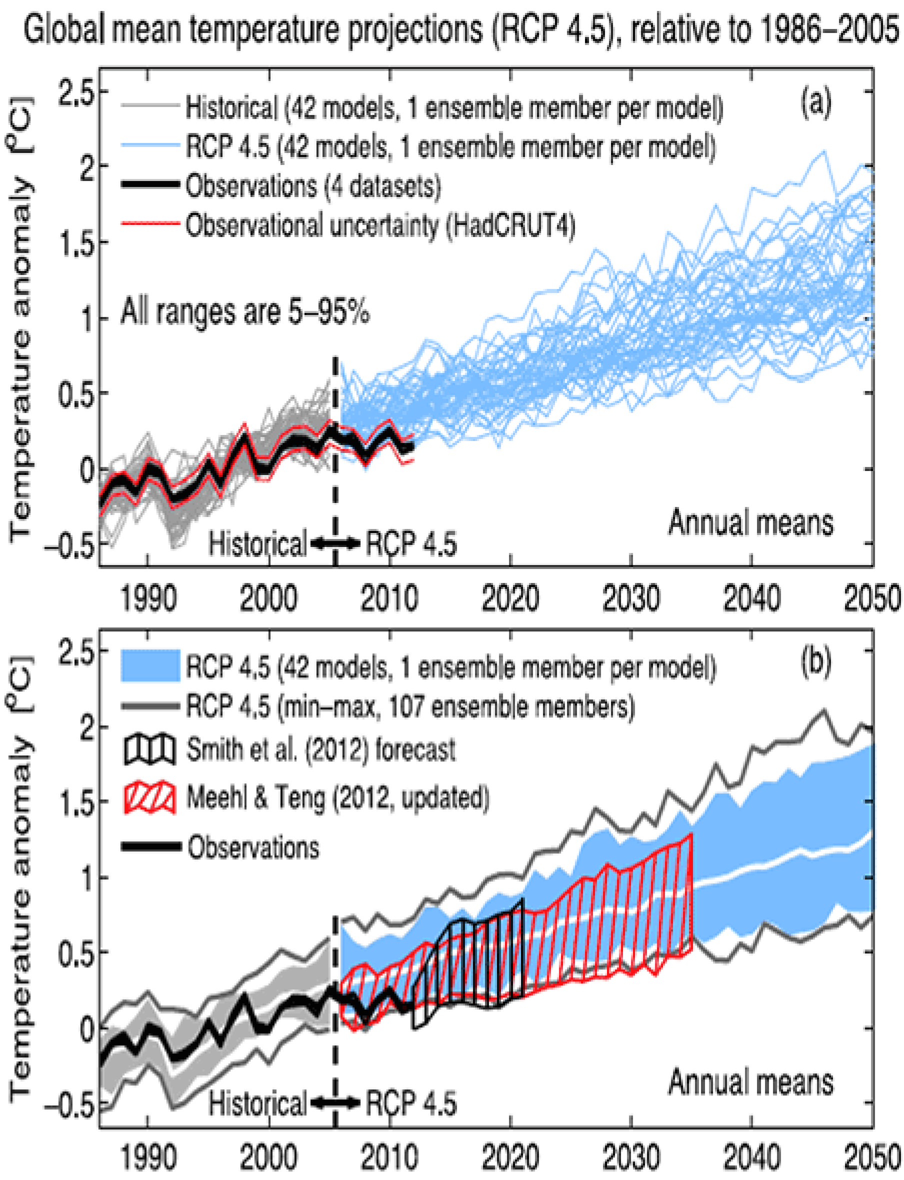

Figure 6. A new way to address the near-term climate change problem with initialized decadal predictions in the AR5. (a) Global surface temperature time series for observations (black) and all ensemble members from the Coupled Model Intercomparison Project (CMIP)5 models (gray for historical, blue for future for Representative Concentration Pathway 4.5) in the AR5; (b) same as (a) except shading represents the range, and the black vertical bars and red vertical bars denote decadal predictions with a subset of the CMIP models initialized in 2015 during the early 2000s slowdown showing the models initialized during the slowdown have less warming than the uninitialized free-running models (blue shading;IPCC, 2013).

This pointed to a new direction for climate science where the time evolution of regional near-term climate had to account for, and models needed to simulate, the combination of naturally occurring internal variability and the response to external forcing, particularly increasing greenhouse gases.

After the criticism of the AR4 regarding perceived estimates of future sea level rise that were too low, the assessment in the AR5 indicated that the high end of the uncertainty range of the highest scenario (RCP8.5) reached 1 m of global sea level rise by 2100 (Figure 7).

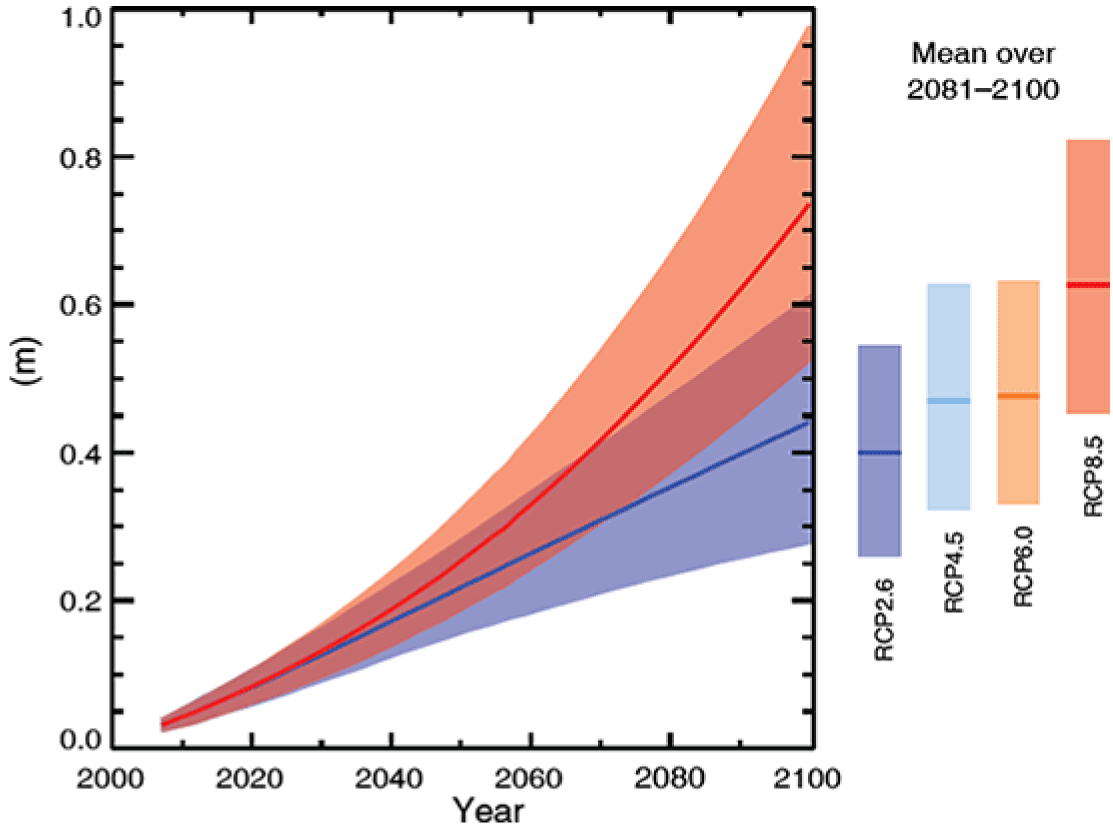

Figure 7. Projected sea level rise from the Coupled Model Intercomparison Project 5 models, time series for the low (Representative Concentration Pathway [RCP]2.6) and high (RCP8.5) scenarios, solid line indicates multi-model mean, shading indicates the range. Means and ranges for all scenarios are at right (IPCC, 2013).

However, the assessment acknowledged that credible interactive ice sheet models were still not able to provide a quantitative estimate of contributions to future sea level rise from melting ice sheets.

Another new science result from the AR5 was the concept that cumulative CO_2 emissions were related to global mean surface temperature change (Figure 8).

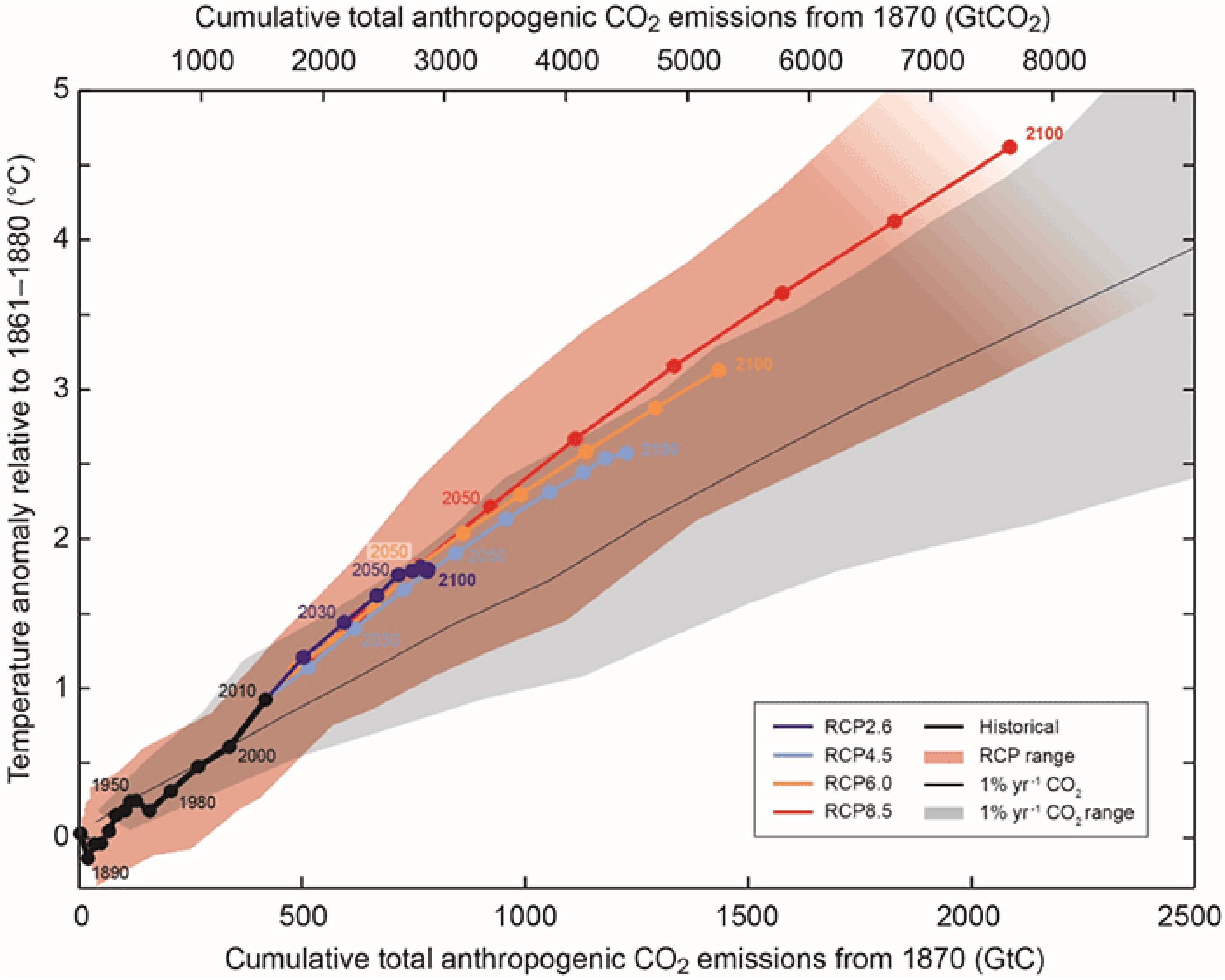

Figure 8. A new way to quantify cumulative CO_2 emissions in relation to global surface temperature change showing a nearly linear relationship between the two, such that higher cumulative CO_2 emissions on the x axis relate directly to higher warming on the y axis. Historical values in black, and the low Representative Concentration Pathway (RCP)2.6 scenario (purple) achieves a warming at 2100 of less than 2.0°C for cumulative CO_2 emissions of about 800 GtC, while the high forcing scenario (RCP8.5) has a warming at 2100 of about 4.7°C with over 2000 GtC (IPCC, 2013).

This provided a new metric for governments to determine how much carbon can still be burned before certain warming thresholds are reached. This developed the new concept of the transient climate response to cumulative emissions of carbon dioxide (i.e., the globally averaged surface temperature change per unit CO_2 emitted), which directly tied a climate change metric (TCR, first assessed in the second assessment report) with cumulative emissions that was further assessed in the IPCC AR6.

For these ideas to gain traction with the governments, there still needed to be an assessment of how strongly the climate science community could tie the burning of fossil fuels to climate change. Recall that the so-called smoking-gun statement from the AR4 was, “Most of the observed increase in globally averaged temperatures since the mid-20th century is very likely due to the observed increase in anthropogenic greenhouse gas concentrations." That was further strengthened, on the basis of new scientific evidence in the AR5, to read, "It is extremely likely that human influence has been the dominant cause of the observed warming since the mid-20th century." This further heightened the sense of urgency from the climate science community and motivated the governments to take steps to mitigate future climate change.

Climate Change Targets: Coupled Model Intercomparison Project phase 6 (CMIP6) and the Sixth Assessment Report (AR6)

Even as the Intergovernmental Panel on Climate Change (IPCC) AR5 process was finishing, the science of climate change was moving rapidly ahead with another new generation of Earth system models (ESMs). More and more research communities wanted to participate in research involving the increasingly complex and sophisticated ESMs. It became clear from the science side that this new generation of ESMs and the many participating research communities would require a new set of coordinated climate change experiments to address an ever-expanding slate of compelling science questions. Attention turned to the possibility of another phase of the Coupled Model Intercomparison Project (CMIP). Motivated by a groundswell of scientific imperatives, planning for CMIP6 started in 2013 for an anticipated start date of 2015 extending through 2020. This planning was motivated by the science and was started fully 3 years before the IPCC AR6 schedule was announced in 2016.

Given the success of the 2006 Aspen Global Change Institute (AGCI) session in formulating CMIP5, the international climate science community convened an AGCI session in 2013 to plan CMIP6. This session brought together about 30 scientists representing a number of scientific communities studying various aspects of climate change and, by definition, scientists who had participated in the three IPCC working groups for the AR5. Participants included ESM modelers and climate scientists, integrated assessment modeling (IAM) modelers, and impacts, adaptation, and vulnerability researchers. Given the experience of the climate science community in CMIP5, the science plan for CMIP6 was formulated somewhat differently (Meehl et al., 2014). CMIP6 would represent a more continuous and distributed activity coordinated by the CMIP Panel under the auspices of the World Climate Research Programme (WCRP) Working Group on Coupled Models. In this way, a relatively large number of "MIPs" would be formed from the various climate science communities interested in having the modeling groups run experiments to address science questions arising in their respective communities regarding climate variability and change. Because this activity was becoming so large, CMIP established protocols for data formats, as well as which data were saved from the model integrations and made available. Additionally, a relatively small number of common experiments were defined that all modeling groups would run to become involved in CMIP (Eyring et al., 2016). These experiments included legacy experiments from earlier CMIP phases that now represented standard metrics of model response to changes of CO_2. These included 1% per year CO_2 increase to doubling to compute the transient climate response, an AMIP simulation (described earlier) with the atmospheric component of the ESM run with time-evolving observed sea surface temperatures, a preindustrial control run with no changes in external forcing, an abrupt $4 \times CO_2$ run from which the equilibrium climate sensitivity could be computed, and a historical simulation from 1850 to 2014 with observed time-evolving greenhouse gases and other forcings. The CMIP Panel fielded proposals for MIPs from the various climate science communities, and those were related to the science foci of CMIP6 (Figure 9).

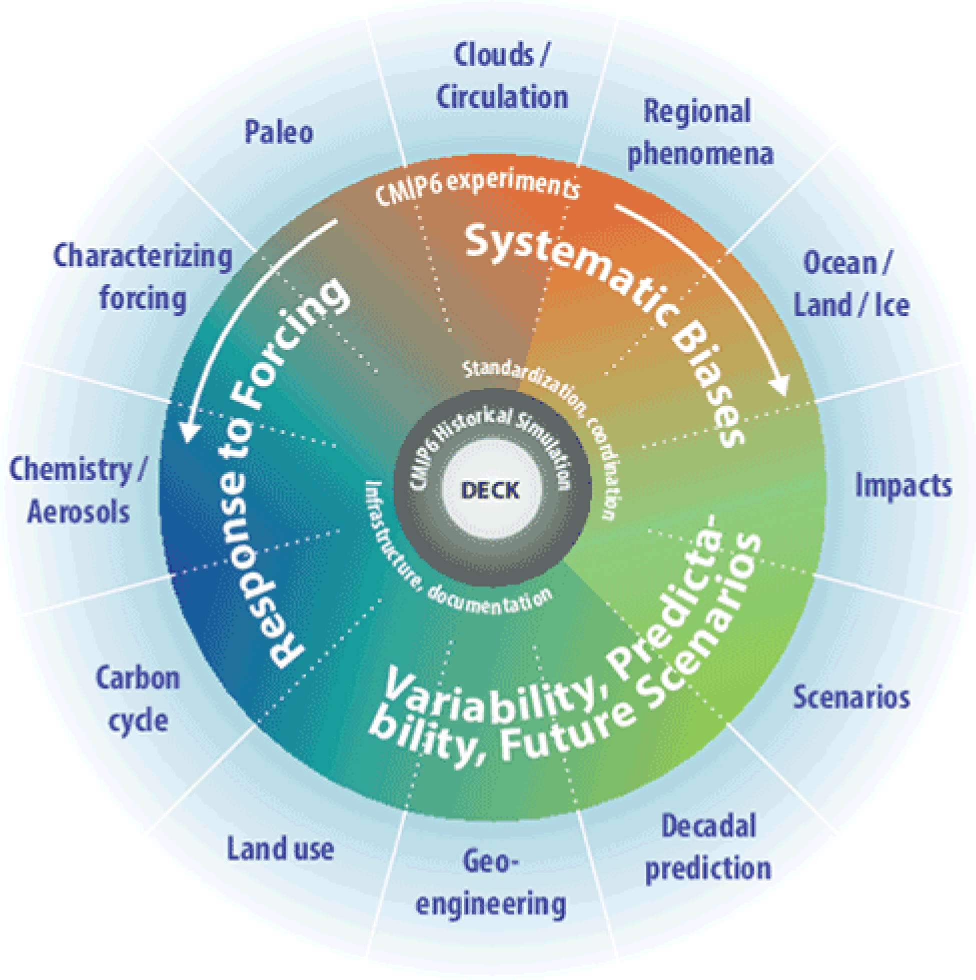

Figure 9. The scientific framework of Coupled Model Intercomparison Project 6, with models that were assessed in the Intergovernmental Panel on Climate Change AR6 (Eyring et al., 2016).

In the end, there were 23 MIPs addressing compelling science questions from the climate science communities in Figure 9. Another new feature was routine evaluation of model data made available by the modeling groups via the Internet-based infrastructure called the Earth System Grid that evolved into the Earth System Grid Federation (ESGF <https://esgf.llnl.gov/>).

As the model simulations progressed and the simulations from the resulting MIPs were ready to be analyzed, scientists around the world were able to access the multi-model data on the ESGF. This resulted in a huge amount of new scientific publications. When the IPCC AR6 timeline was announced in 2016, CMIP6 and new climate science results were well on their way to producing a staggering amount of new science to be assessed in the AR6.

Another significant development that occurred as part of post-AR5 science was the formulation by the IAM groups of a new set of emission scenarios called the Shared Socioeconomic Pathway (SSP) scenarios that were part of a CMIP6 MIP called “ScenarioMIP” (O’Neill et al., 2016). A number of these scenarios involved various levels of mitigation with an eye on warming targets that could inform policy negotiations. The results from CMIP3 with the SRES scenarios showed that the Representative Concentration Pathway (RCP)2.6 scenario could keep global warming below 2°C (Figure 5). However, government negotiations at the UN Framework Convention on Climate Change meeting in Paris in late 2015 aspired to keeping warming well below 2°C above preindustrial, with a final agreed-upon target of 1.5°C above preindustrial. Consequently, as an example of the interactions in the IPCC context between the science community and the governments, emission pathways to achieve the 1.5°C target were formulated (Rogelj et al., 2018). One of the SSP scenarios designed to achieve this target (SSP1–1.9) was run by the Earth system modeling groups (O’Neill et al., 2016).

While the final stages of the AR6 assessment process were complicated by the onset of the worldwide COVID pandemic in 2020, the Working Group 1 assessment was finalized in 2021 (IPCC, 2021). As always, there was interest in the smoking-gun statement. It had been strengthened through the evolution of the previous assessments as the science became increasingly more conclusive. Recall that the statement in the AR5 was, “It is extremely likely that human influence has been the dominant cause of the observed warming since the mid-20th century.” The statement got even stronger in the AR6: “It is unequivocal that human influence has warmed the atmosphere, ocean and land.”

A range of warming by the end of the 21st century was provided by CMIP6 model simulations using the SSP scenarios (Figure 10a).

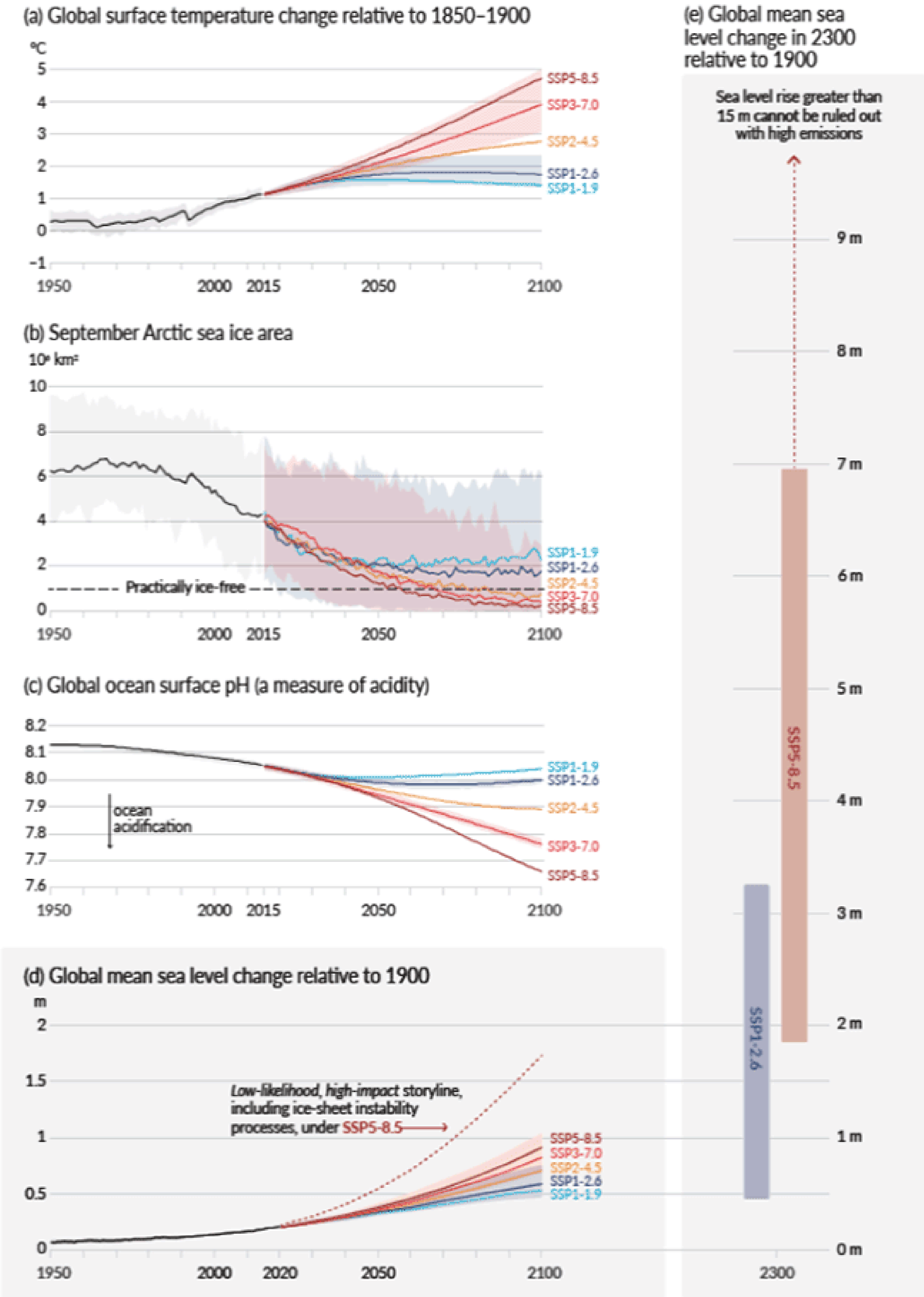
(a) Global surface temperature change relative to 1850–1900
°C
5
4
3
2
1
0
-1
SSP5-8.5
SSP3-7.0
SSP2-4.5
SSP1-2.6
SSP1-1.9
1950
2000
2015
2050
2100
(b) September Arctic sea ice area
10⁶ km²
10
8
6
4
2
0
Practically ice-free
SSP1-1.9
SSP1-2.6
SSP2-4.5
SSP3-7.0
SSP5-8.5
1950
2000
2015
2050
2100
(c) Global ocean surface pH (a measure of acidity)
8.2
8.1
8.0
7.9
7.8
7.7
7.6
ocean acidification
SSP1-1.9
SSP1-2.6
SSP2-4.5
SSP3-7.0
SSP5-8.5
1950
2000
2015
2050
2100
(d) Global mean sea level change relative to 1900
m
2
1.5
1
0.5
0
Low-likelihood, high-impact storyline, including ice-sheet instability processes, under SSP5-8.5
SSP5-8.5
SSP3-7.0
SSP2-4.5
SSP1-2.6
SSP1-1.9
1950
2000
2020
2050
2100
(e) Global mean sea level change in 2300 relative to 1900
Sea level rise greater than 15 m cannot be ruled out with high emissions
9 m
8 m
7 m
6 m
5 m
4 m
3 m
2 m
1 m
0 m
SSP5-8.5
SSP1-2.6
2300

Figure 10. Results from the Coupled Model Intercomparison Project (CMIP)6 models assessed in the AR6, updating a comparable plot for the CMIP5 models in the AR5 in Figure 5. (a) Global surface temperature change from the CMIP6 models for the Shared Socioeconomic Pathway (SSP) scenarios; (b) estimates from the CMIP6 models as to when the Arctic could become ice free in September; (c) estimates from the CMIP6 models regarding the magnitude of future ocean acidification; (d) estimates from the CMIP6 models regarding possible future global sea level rise for the SSP scenarios that have mostly contributions from thermal expansion, with the dotted line indicating a possible greater increase of sea level rise if the Greenland and Antarctic ice sheets destabilize; and, panel (e) showing possible sea level rise for the year 2300 relative to 1900 (IPCC, 2021).

Attendant estimates of when the Arctic could become ice free in September (Figure 10b) with the high forcing scenario (SSP5−8.5) indicated this could occur around 2055, just a bit later than the CMIP5 models for RCP8.5 of about 2050 (Figure 5b). Levels of ocean acidification from the CMIP6 ESMs scaled roughly with warming in the respective scenarios (Figure 10c). The previously contentious issue of sea level rise now computed from the SSP scenarios did not look much different from the RCP scenarios, with the high end of the highest scenario (SSP5−8.5) showing about 1 m of sea level rise by 2100 (Figure 10d) and much larger values by 2300 (Figure 10e). But a speculative scenario that involved ice sheet instability, which was still not well quantified, estimated a possible sea level rise of over 1.5 m by 2100 (Figure 10d), pointing to the present limits of human understanding of ice sheet processes and our ability to model and credibly project changes in those processes.

The now standard geographic maps of surface temperature change from the CMIP6 ESMs in the AR6 (Figure 11) showed comparable patterns of warming to what the very early coupled climate models showed in the First Assessment Report (Figure 1).

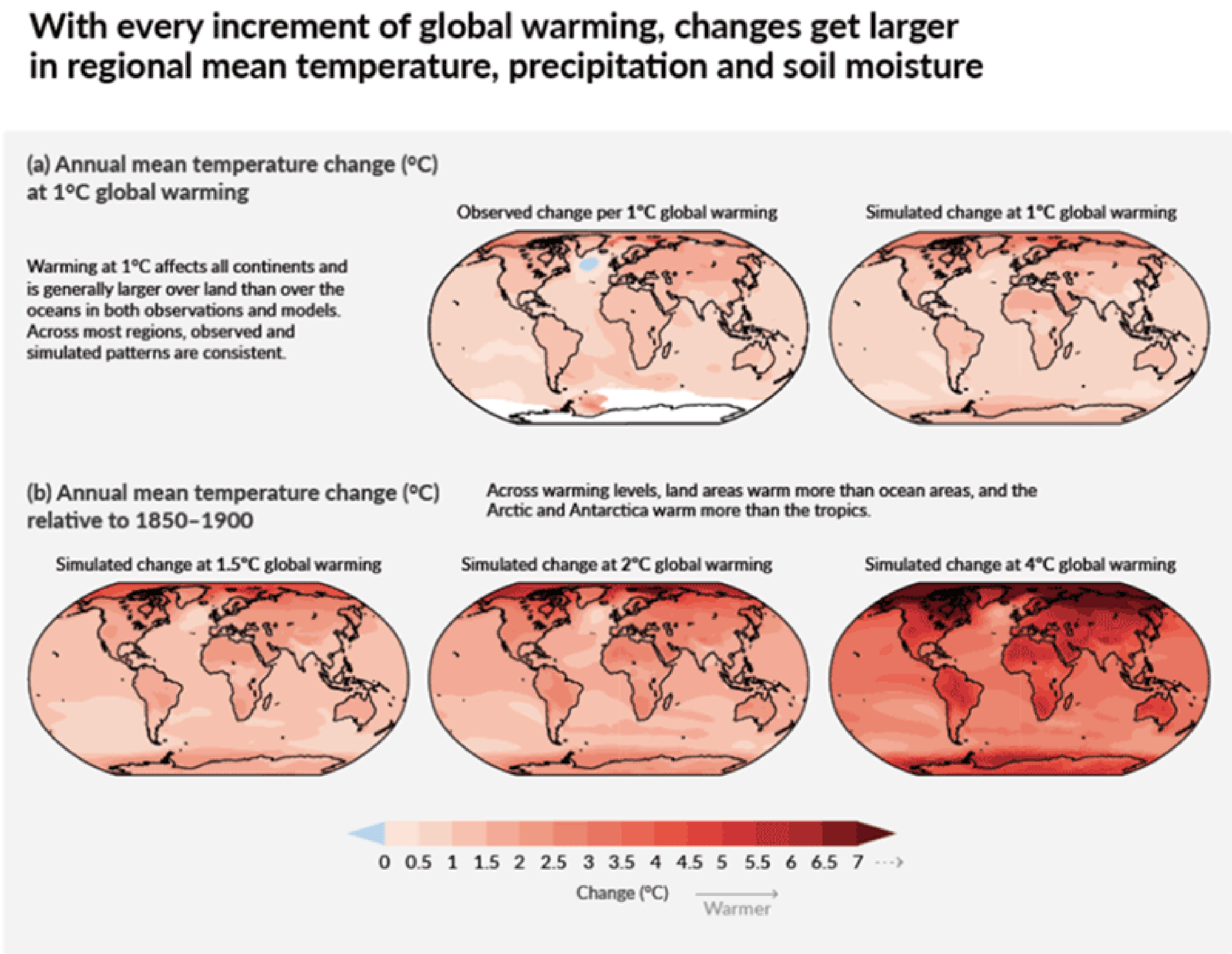

Figure 11. The depiction of geographical patterns of surface temperature change from the Coupled Model Intercomparison Project (CMIP)6 models assessed in the Intergovernmental Panel on Climate Change AR6 (IPCC, 2021), with anomalies plotted for different levels of warming. Note similarities in the patterns compared to the first generation of coupled climate models in the early 1990s in Figure 1, and the 2001 vintage CMIP2 models in the Third Assessment Report (Figure 3).

As noted earlier, this is reassuring in that this kind of consistency of a model result across many generations of models with ever-increasing complexity and realism indicates the very fundamental nature of the Earth system response to increases of greenhouse gases. It shows that these patterns, starting with the First Assessment Report and when compared to observations, indicate that these climate changes are undeniably emerging in the real climate system. This verifies the credibility of the ESMs used to predict many aspects of climate.

Finally, in an example of a policy relevant science result, the AR5 conclusion that there is a near-linear relationship between cumulative CO_2 emissions and global warming was confirmed in the CMIP6 models in the AR6 across all the SSP scenarios (Figure 12).

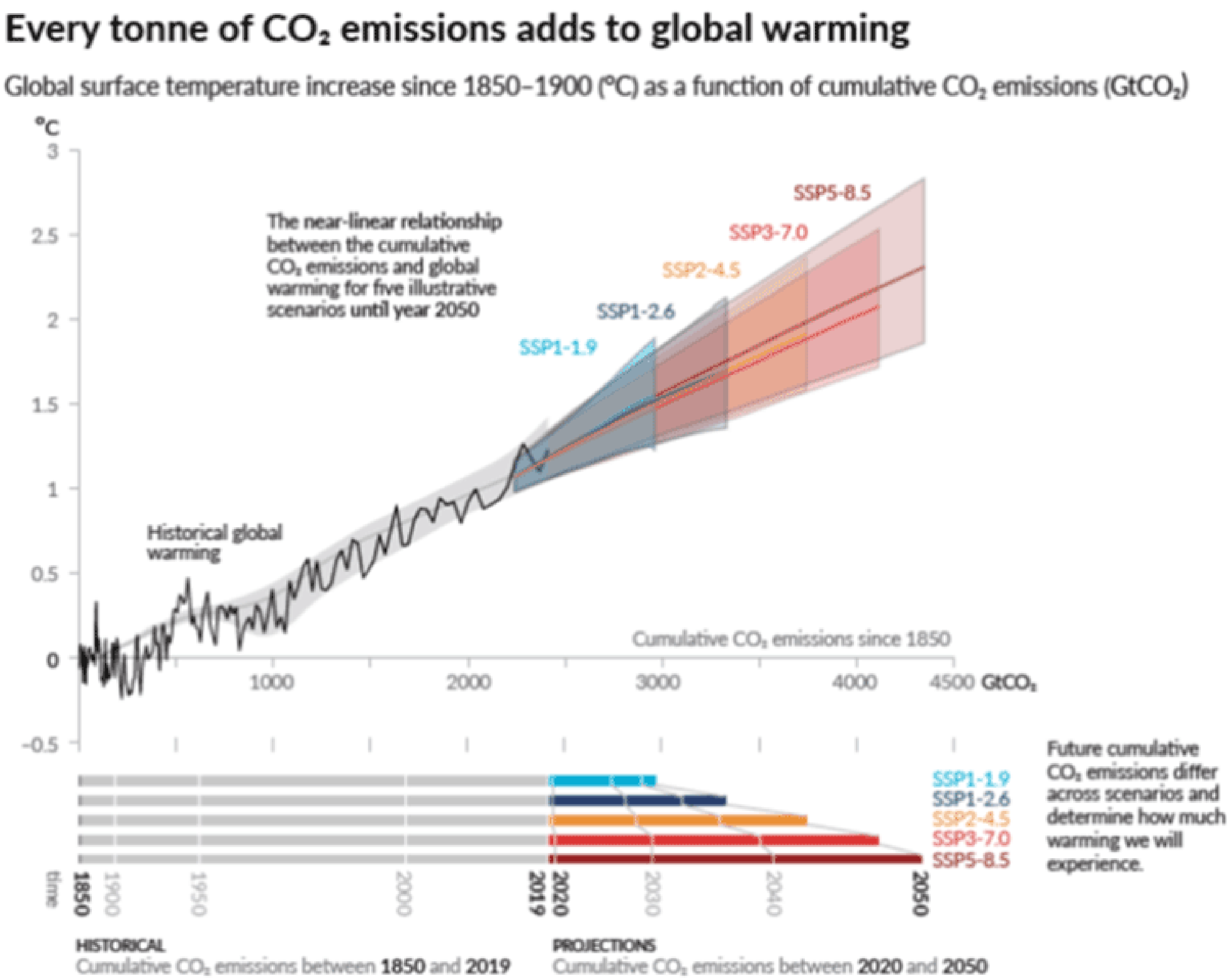

Figure 12. Confirmation in the AR6 of the result in the AR5 (Figure 8) that there is a near-linear relationship between the cumulative CO_2 and global warming across all the Shared Socioeconomic Pathway scenarios (IPCC, 2021).

Thus, for different levels of global warming, this allows a way to estimate how much carbon there is left to burn before reaching various warming levels. Given the focus on warming targets in the Paris Agreement, this kind of science result is very informative for policy negotiations and provides an urgency to those agreements.

The Future of Climate Science and the IPCC

After the conclusion of every Intergovernmental Panel on Climate Change (IPCC) assessment, there are questions as to whether there should be another. As the field of climate science has expanded, so has the scope of an IPCC assessment, thus making such an exercise involving hundreds of scientists and many hundreds of scientific papers more and more unwieldy. But the governments have always liked the IPCC assessments because they provide the current state of knowledge of the climate change problem to motivate and justify ever more serious efforts on the parts of the governments to do something about climate change. This came to fruition in the Paris Agreement of 2015, when governments agreed to emission targets to attempt to hold global warming to 1.5°C above preindustrial levels. In spite of the best intentions of the governments, these targets have been extremely difficult to achieve, and it remains to be seen whether the increasingly urgent results from climate science will move the governments to stick to their commitments with regard to emission targets.

An example of what the future of the IPCC may look like occurred after the AR5. There was a feeling that another full assessment may not be needed, but rather a series of special reports targeted at specific aspects of the science most relevant for policy negotiations could be performed. The discussion started out as "either-or" on the science side and ended up as "both" from the government side. Therefore, since the AR5 in 2013 there have been several special reports that were followed by the full assessment published in 2021. One was on global warming targeting 1.5°C (Masson-Delmotte et al., 2018), another was on the ocean and cryosphere in a changing climate (Pörtner et al., 2019), and a third was on climate change and land (Shukla et al., 2019). IPCC has also convened a number of workshops and expert meetings on various hot topics in climate change science, and these reports and the full collection of the IPCC assessments is available at IPCC Library *<https://www.ipcc.ch/library/>*. Also note that the focus of the present article has been on the physical climate science represented in the Working Group 1 reports, but the reader should also be aware of comparable assessments from the other two working groups, and those are also available at IPCC Library *<https://www.ipcc.ch/library/>*.

From the climate science side, as was noted, the IPCC assessment cycle has provided compelling motivation to move rapidly forward on the science of climate change. This drive arose from the urgent need to learn as much as possible about not only the functioning of the climate system, but also how it may respond to increases in human-produced greenhouse gases, arguably one of the biggest scientific challenges facing society today. But the IPCC assessments could only gauge the state of human knowledge of climate change from the published scientific literature, and various coordination mechanisms have been formulated on the science side to advance the science as rapidly as possible. One crucial element that has been instrumental in coordinating and pushing forward Earth system modeling has been the Coupled Model Intercomparison Project (CMIP). The IPCC does not run CMIP, but rather it is an activity organized by climate scientists within the World Climate Research Programme (WCRP) to structure the science to make advances as rapidly as possible. Thus, the science produced through CMIP is critical to the IPCC assessments because the CMIP models and experiments are the state-of-the-science upon which rests much of human knowledge of how the system works as represented by scientific publications arising from the analysis of the CMIP experiments.

The IPCC assessments have also proved useful as a periodic stock-take of climate science. Many a graduate student has been pointed to the latest IPCC assessment to come up to speed on where the field of climate science is right now. The scientific literature assessed in each report provides an encyclopedic compilation of the latest papers being published in the field. A look at the IPCC assessments through time, starting with the first report in 1990, provides a unique history of a scientific field of endeavor that probably does not exist at that exhaustive a level for any other area of science.

The concept has been developed in this article about the symbiotic relationship between climate science and the IPCC. The foundation of the IPCC assessments depends on state-of-the-art scientific results, and climate scientists performing that research are fully mindful of the benefits of having cutting-edge research results appearing in the high-profile IPCC assessments. That provides additional motivation for the scientists to move forward as rapidly as possible to advance human knowledge of climate variability and change. Of course, most climate scientists are motivated by the compelling nature of the human-caused climate change problem itself, independent of the IPCC assessments. Climate scientists view this problem as of paramount importance for the future of planet Earth and gain motivation from the urgent nature of trying to understand all the facets of the problem.

Going forward, as the impacts of human-caused climate change become increasingly severe, and the pressure mounts on the governments of the world to do something about it, climate science will continue to advance the state of human knowledge of climate variability and change to inform and motivate policy decisions to mitigate the problem. Climate science has seen an explosive growth and now includes many scientific communities that study various aspects of the Earth system. This expansion will continue to include additional communities and scientists as the climate models become more complex and the field of assessing the impacts of climate change expands. Meanwhile, the IPCC will likely continue to evolve as well, with additional targeted special reports (such as one on climate change and cities already being discussed for the seventh assessment cycle). Governments will continue to want to know the latest science to inform their decisions, and the future will be one of a combination of mitigation measures to limit GHG emissions, and adaptation to the climate change that human societies cannot or will not be able to mitigate. Therefore, there is even more pressure on the climate science community, and the Earth system–modeling community in particular, to provide quantitative and reliable information on the time evolution of regional climate change over the next years to decades, because those are the conditions to which human societies must adapt. This is a huge scientific challenge, and the stakes could not be higher. The climate science community is already rising to this challenge, and the IPCC assessments and special reports will continue to provide the latest scientific information to governments to inform the decisions that will determine the fate of our planet.

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